

AuraOS Paper IX: Objective-Native Capability Commons and Proof-Carrying Contribution Economies

Portable Cognitive Capsules, Proprietary Capability Execution, Developer Arenas, Attestation Lineage, Aura Places, Convention Arenas, Participatory R&D, Simulation, and Business Incubation

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Abstract

This paper continues the AuraOS defensive-publication sequence after Paper VIII, which disclosed claims N31-N50 for evidence-ordered relational Arenas, compiled relationship intelligence, verified engineering, governed spatial projection, continuity, and atomic publication. Paper IX discloses a further architectural stratum: **objective-native compositional computing built around persistent capabilities, ephemeral applications, proof-carrying contribution, portable personal cognition, federated Commons infrastructure, and governed human/AI collaboration.**

The central disclosed operating model is:

```
human / organization / community objective
|
v
personal semantics + canonical intent + constraints
|
v
minimum-sufficient evidence and capability discovery
|
v
rights / authority / identity admission
|
v
typed composition plan or Arena Recipe
|
v
ephemeral Arena
|
+--> humans
+--> AI workers / personal SLMs
+--> open or proprietary capability packages
+--> rule packs / data / simulations / services
+--> contributed compute / scientific facilities
|
v
verification + semantic-gate execution receipts
|
v
Attestation / Provenance DAG
```

```

      |
      +---> causal contribution analysis
      +---> human / institutional responsibility attestation
      |
      v
canonical-owner disposition
      |
      v
attribution / entitlement / economic settlement
      |
      v
explicit promotion of durable value
      |
      v
deterministic dissolution of temporary state

```

Paper IX formalizes **thirty-seven new combination-scoped defensive prior-art declarations, N51-N87**. They cover: an objective-native minimum-sufficient Arena compiler; hierarchical source hydration; persistent Capability Packages; Arena Recipes; explicit promotion and dissolution; federated Aura Commons; executable rights; proprietary-code participation without mandatory source disclosure; semantic-gate receipts and lazy provenance; durable agent identity integrated with bounded authority; meaningful-use attribution and continuous optimization economics; proof-carrying Developer Arena parallelism; Personal Cognitive Capsules and portable personal SLMs; semantic translation and privacy membranes; governed recursive harness learning; intent-native manifestation and spatial code breadboarding; Aura Places and relationship-first social commerce; Convention Arenas; dual reactive/proactive discovery; an Open Discovery Foundry; physics/digital-twin simulation; bounded population-behavior simulation; a business-incubation Arena; sovereign cross-domain federation with minimized independent review; participatory proof-carrying Scientific Arenas; and a compounding Scientific Capability Commons in which verified reusable improvements can propagate across future compatible research objectives while preserving provenance, authority boundaries, and explicit economic settlement contracts.

The disclosures are intentionally **combination-scoped**. Existing primitives such as FST/WFST routing, vector symbolic representations, capability security, OI DC, DNS, Verifiable Credentials, transparency logs, Merkle DAGs, social federation protocols, WebAssembly, Gaussian splatting, agent-based modeling, digital twins, marketplace accounting, and trusted execution are not claimed in isolation. The asserted prior art is the specific composition, evidence ordering, authority separation, lifecycle, provenance, privacy, portability, and economic feedback disclosed here.

1. Status, Scope, and Defensive-Publication Purpose

This document is a technical defensive publication and reference architecture. It is intended to place the disclosed combinations and embodiments into the public technical record. It is **not** a legal opinion, patentability determination, freedom-to-operate opinion, safety certification, financial/legal compliance certification, or guarantee against later patent assertions.

The paper distinguishes three implementation-status classes:

R - Repository-backed substrate. Implemented or directly evidenced in the AuraOS repository at the referenced snapshot; used here as inherited implementation foundation.

P - Refactor-specified architecture. Defined in current numbered refactor plans or governed harness specifications but not necessarily merged; disclosed here as an engineering design and planned integration.

E - Enabling reference embodiment. Fully described so an engineer can implement the combination even when downstream deployment is future-gated; used to establish the architecture without falsely claiming deployment.

The repository anchor used for the contemporary baseline is:

```

Repository: dallascourchene-commits/AuraOS
Branch:     main

```

Commit: 9c04a1efa57461a6078acb9f3b569766cbd2ab24
Commit date: 2026-08-06 UTC
Commit note: PR1: Add intent-native spatial workspace contracts (#255)

Paper VIII used an earlier July 20, 2026 repository snapshot. Paper IX does not rewrite N31-N50; it incorporates them as substrate.

2. Continuity With Papers I-VIII

AuraOS maintains a defensive-publication stack in which Papers I-VII disclosed claims N1-N30 and Paper VIII disclosed N31-N50.

The relevant inherited architectural rules are:

1. **Planning proposes. Governance authorizes. Verification proves.**
2. Exact source, contracts, domain records, and verifier evidence outrank advisory semantic or visual representations.
3. Hard admission guards execute before soft ranking.
4. Each consequential truth has a canonical owner; projections may reduce authority but cannot enlarge it.
5. Arenas are bounded and leased.
6. Model output, visual topology, VSA similarity, research, simulation, consensus, and learned heuristics cannot manufacture authority.
7. External workers receive bounded slices and return proposal/evidence state.
8. Learning is proposal-only until independently promoted.
9. Spatial presentation is not domain truth.
10. Publication and merge remain separately authorized.

Paper VIII expressly left module marketplaces, intent-compiled consumer application fabrics, broader federation, and related product directions as future programs. Paper IX supplies the **enabling disclosure for that next stratum**.

3. Paper IX Core Thesis

Conventional computing ordinarily starts with:

Which application should I open?

The disclosed Aura model begins with:

What is the person, organization, or community trying to accomplish?

The application becomes a temporary compilation result rather than the permanent unit of computing.

A simplified reference formulation is:

```
Arena_q = Compile(  
    Objective_q,  
    Semantics_q,  
    Evidence_q,  
    Constraints_q,  
    Capabilities_q,  
    Rights_q,  
    Authority_q,  
    Identity_q,  
    Context_q,  
    Device_q,  
    Budget_q,  
    ProofObligations_q,  
    Completion_q,
```

```
Dissolution_q
)
```

The permanent system is therefore not a catalogue of monolithic applications. It is a governed substrate containing persistent capabilities, recipes, relationships, identity, rights, provenance, evidence, and reusable knowledge.

PERSISTENT VALUE

```
Capability Packages
Arena Recipes
Places
Rights / Entitlements
Provenance / Attestation
Verified Knowledge
Relationships
Compatibility Evidence
Contributor Lineage
```

```
+
|
v
```

EPHEMERAL COMPOSITION

```
objective-specific state
activated capability leases
temporary workers
generated layouts
proposal branches
renderer state
temporary secrets
task context
```

```
=
|
v
```

OBJECTIVE-SPECIFIC ARENA

4. One Constitutional Plane, Many Bounded Surfaces

Paper IX does not introduce a second truth, routing, memory, verification, authority, identity, payment, or governance plane.

The disclosed components are adapters, contracts, registries, projections, and execution surfaces inside or around the existing constitutional architecture.

A useful authority relation is:

```
authority(projection) <= authority(canonical owner)
authority(child worker) <= authority(parent worker)
authority(recipe) = 0 unless rebound to current rights/leases
authority(simulation) = advisory only
authority(rendered Place) = 0 for canonical mutation
authority(payment record) != authority to change domain truth
```

No semantic score, market popularity, credential, signature, agent identity, or economic payment overrides a failed hard guard.

5. Objective-Native Compilation and Minimum-Sufficient Hydration

5.1 The compiler

The reference compiler accepts not only a positive objective but also explicit negative intent:

```
objective: "Evaluate a proposed capability integration"
do:
  - preserve the canonical interface
  - measure latency and security impact
  - retain upstream attribution
do_not:
  - expose proprietary source
  - expand network access
  - weaken verifier requirements
  - modify production automatically
privacy:
  disclose_only: "contract and output proof"
budget:
  compute: bounded
  time: bounded
authority:
  proposal_only: true
```

This bilateral intent is compiled into work-unit contracts, capability admission, evidence requests, and invalid transitions.

5.2 Evidence hierarchy

Aura does not obtain efficiency by deleting evidence. It obtains efficiency by hydrating exact evidence at the point of consequence.

```
objective
|
v
cheap semantic / topological orientation
|
v
relationship cluster
|
v
capability / object / policy / time region
|
v
symbol / record / parameter set
|
v
exact source / frame / authoritative record
```

The same hierarchy supports semantic zoom in a spatial or conventional UI.

Examples:

- code: architecture -> module -> function -> AST/source;
- Place: Place manifest -> room -> object -> high-detail asset;
- video: transcript -> time segment -> key frames -> exact frames;
- physics: phenomenon -> model family -> parameters -> empirical data;
- governance: authority class -> applicable rule -> exact authoritative text;
- personal memory: semantic primitive -> episode -> evidence;
- marketplace: objective -> capability role -> package -> execution contract.

6. Capability Packages and the Proprietary-Code Commons

6.1 Capability Package

A Capability Package is a persistent versioned unit of specialist functionality.

Illustrative manifest:

```
capability_id: cap.example.validator.v4
publisher: publisher.example
version: 4.2.1
```

```
contract:
  inputs:
    - CapabilityIntent
    - ArenaManifest
  outputs:
    - VerifiedCapabilityRequest
  effects:
    - NONE
  deterministic: true
```

```
rights:
  required:
    - execute
  source_visibility: PRIVATE
```

```
runtime:
  modes:
    - SIGNED_VENDOR_RUNTIME
    - TEE_ATTESTED
  network: denied
  filesystem: leased
```

```
evidence:
  verifier_pack: vp.example.validator.v2
  reproducibility_class: TEE_ATTESTED
  provenance_root: sha256:...
```

```
economics:
  licence: commercial
  pricing_mode: per_verified_success
  upstream_attribution:
    - dependency.example
```

```
cleanup:
  mandatory: true
  receipt_required: true
```

6.2 Evidence modes for proprietary execution

The Commons does not pretend all packages provide equal transparency.

- **OPEN_SOURCE_REPRODUCIBLE** - public source/build; reproducible builds, tests, and signatures provide the broadest technical inspectability.
- **SOURCE_ESCROW_REPRODUCIBLE** - source is disclosed only to an authorized verifier/escrow process; independent build and test evidence can support confidentiality-sensitive admission.

- **TEE_ATTESTED** - the implementation remains private inside an attested execution image; hardware/runtime attestation and output proof provide a different assurance class.
- **SIGNED_VENDOR_RUNTIME** - the vendor controls the binary/runtime; signatures, exact version identity, contract tests, and monitoring provide evidence but lower source inspectability.
- **REMOTE_SERVICE_ONLY** - no local implementation is visible; endpoint identity, contracts, receipts, and behavior tests form the narrowest assurance mode and may be prohibited for high-risk work.

A package may be economically useful without being admitted to every risk class.

6.3 Private source and public attribution are compatible

A meaningful-use receipt does not need the source code.

It may bind:

```
package identity
package version
implementation digest / attestation
input class or hash
authorized purpose
execution result
output hash
verifier result
downstream consumption
economic attribution identity
```

The publisher can therefore receive credit or payment while the user does not disclose private objective data beyond the allowed receipt and the publisher does not disclose source beyond its execution/evidence mode.

7. Arena Recipes

A recipe is the reusable pattern, not the implementation.

Example:

```
recipe: SmallCommercialRenovationEvaluation
roles:
- building_code
- plan_parser
- cost_estimator
- schedule_engine
- safety_rule_pack
- spatial_projector
- human_review
proof:
- all hard blockers evaluated
- cost assumptions labeled
- source/code versions pinned
- final decision external
dissolution:
- release temporary project slices
- retain authorized receipt roots
```

At runtime each role is resolved against current capability versions, policy, rights, data location, cost, and evidence.

This allows a recipe creator to contribute valuable composition knowledge independently of the developers who created the underlying capabilities.

8. Aura Commons

8.1 Commons as governed substrate

The Commons may coordinate:

- capability and recipe discovery;
- package identities;
- publisher/maintainer records;
- licences;
- entitlements;
- rights;
- provenance;
- verification evidence;
- reputation derived from verified outcomes;
- performance evidence;
- security/revocation state;
- bounties and demand signals;
- payment references;
- community registries;
- Places;
- assets and procedural generators;
- export/migration;
- dispute/appeal procedures.

It should support both public and proprietary economic participation.

8.2 Federated registries

```
Global Commons Registry -----+
Enterprise Private Registry ----+
Nation / Community Registry ----> objective-scoped resolution
Professional Registry -----+
Local Offline Registry -----+
```

No one registry becomes the universal truth owner.

A Nation-controlled registry can restrict culturally governed material while still publishing compatibility information that allows an authorized Arena to know that a capability exists.

8.3 Anti-capture properties

Reference requirements include:

- portable manifests;
- portable Places;
- exportable receipts and entitlements;
- transparent ranking evidence;
- no forced exclusivity by default;
- dependency portability;
- private and federated registries;
- user-selectable discovery priorities;
- limits on self-preferencing;
- community appeals;
- revocation/recovery;
- licence-appropriate source/evidence access.

9. Executable Rights and Entitlements

Rights must change feasible action rather than act as labels.

Illustrative object:

```
right_id: right-2026-081
principal: dnsid:worker.vendor.example
resource: cap.example.physics_solver.v3
purpose: "bounded simulation"
effects:
  - read_input_model
  - execute_solver
  - return_simulation_result
prohibited:
  - modify_canonical_project
  - procure_material
  - publish_external_claim
delegation:
  allowed: false
expires: 2026-08-07T21:00:00Z
revocable: true
evidence_required:
  - execution_receipt
  - simulation_campaign_receipt
```

A marketplace purchase grants only the entitlement expressed by the purchased right. It does not transfer domain authority.

10. Identity, Attestation, Provenance, and Responsibility

10.1 End-to-end accountability chain

```
DNSid / external durable agent anchor
|
v
Aura principal identity
|
v
objective + authority lease
|
v
worker / runtime / model identity
|
v
semantic-gate execution receipt
|
v
Attestation DAG
|
v
causal / contribution analysis
|
v
verification result
|
v
```

```

human / institutional responsibility declaration
    |
    v
Commons / contract / governance consequence

```

Each layer answers a different question.

1. **External durable identity:** Whose agent is this, and which accountable entity anchors it?
2. **Aura authority:** What was it allowed to do for this objective?
3. **Execution receipt:** What did it actually do?
4. **Provenance DAG:** What did the result derive from?
5. **Causal attribution:** What likely contributed materially?
6. **Verifier:** Which bounded predicates were actually checked?
7. **Responsibility attestation:** Who reviewed the artifact and stands behind which declaration?
8. **Governance/economic layer:** What consequence, credit, payment, remedy, or appeal follows?

10.2 DNSid boundary

As of June 29, 2026, the IETF DNSid Internet-Draft describes a minimal DNS-anchored durable identity primitive for AI agents. It binds an agent FQDN to an accountable entity and points to key, status, and lifecycle information while explicitly leaving general authorization, policy enforcement, trust/reputation, and runtime execution to consuming systems.

Aura can therefore consume DNSid-like identity as an **external accountability anchor** rather than building a competing global identity plane.

10.3 Machine and human attestation remain distinct

A machine receipt may say:

```

Worker A produced output X
from input digests B/C
under lease L and policy P.
Verifier V returned PASS for predicates Q.

```

A human/institutional responsibility declaration may say:

```

Named principal reviewed artifact X
and stands behind declaration Y
at time T.

```

The second may be implemented by an external system such as an Opus-Facio-like accountability service. Paper IX does not claim that external mechanism's specific algorithm.

11. Semantic-Gate Receipts and Lazy Provenance

11.1 Gate model

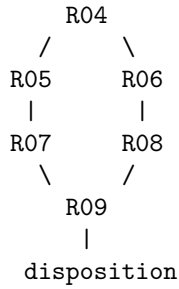
```

G0  objective admitted
G1  intent normalized
G2  constraints admitted
G3  capability selected
G4  graph/plan admitted
G5  authority granted
G6  action opened
G7  material state changed
G8  verification executed
G9  result admitted/rejected
G10 human/community disposition
G11 persistence/publication/closure

```

Millions of internal operations can occur between gates without generating millions of provenance records.

11.2 Receipt DAG



11.3 Receipt frontier

Active state may retain only:

```

run_id: ...
current_state_hash: ...
current_receipt: R09
receipt_frontier: [R07, R08]
checkpoint_root: ...

```

When a reviewer asks “why did this interface exist?”, only the relevant parent receipts and evidence are hydrated.

12. Developer Arena and Massive Verified Collaboration

12.1 Atomic Work Unit

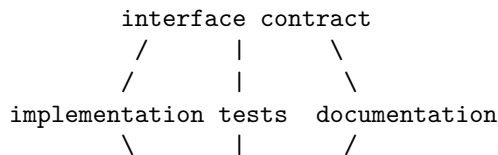
```

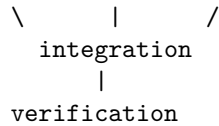
work_unit:
  objective: ...
  canonical_semantics: [...]
  exact_source_state: ...
  allowed_mutation_scope: [...]
  dependencies: [...]
  interface_contract: ...
  do: [...]
  do_not: [...]
  authority: proposal_only
  budget: ...
  proof_obligations: [...]
  expected_outputs: [...]
  failure_classes: [...]
  retry_rules: ...
  parent_state_hashes: [...]
  attribution_terms: ...
  completion_receipt: required

```

The worker can be human or machine.

12.2 Critical-path compilation





The architecture seeks to convert avoidably sequential dependencies into safely parallel work.

The target is not:

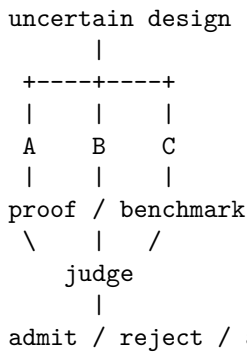
spawn as many agents as possible.

It is:

minimize verified objective completion latency.

12.3 Speculative branches

When a critical uncertainty is cheap enough to explore:



Every branch remains attributable and disposable.

12.4 Integration

Outputs are integrated according to the verified dependency DAG, not the order of chat messages.

Changed upstream assumptions invalidate only affected descendants when the graph proves other work remains reusable.

13. Contribution Attribution and the Self-Feeding Economy

13.1 Meaningful-use evidence

The economic unit is a meaningful verified contribution to an accepted objective.

A reference contribution vector can include:

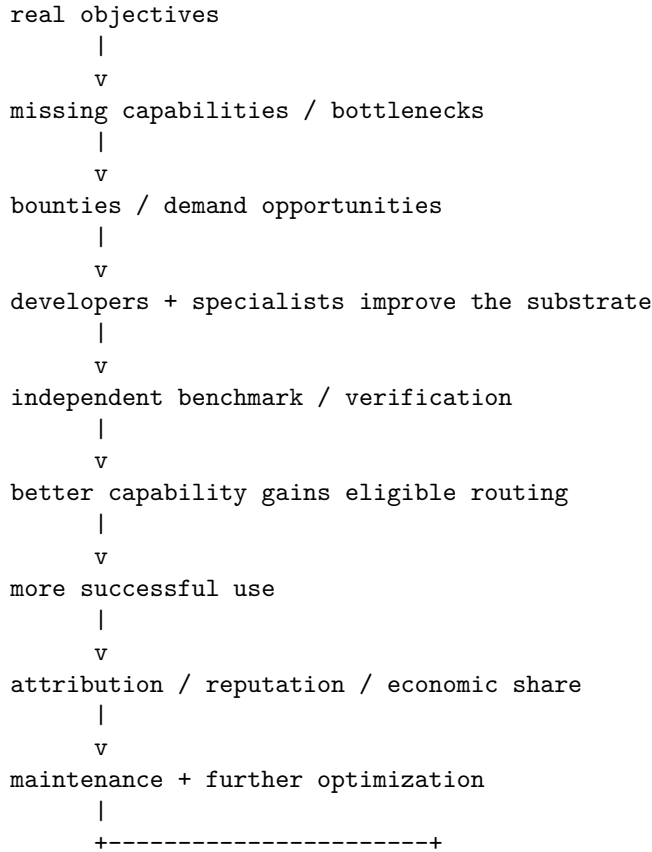
```

executed?
verified?
output survived?
downstream consumed?
prevented failure?
user/canonical owner accepted?
upstream enabled?
superseded?
maintenance/security role?
quality delta?
latency/compute/bandwidth delta?
counterfactual marginal effect?

```

No one field is a universal payout formula.

13.2 Economic flywheel



This can reward invisible infrastructure such as parsers, adapters, accessibility tools, security checks, compression formats, and maintainers.

13.3 Candidate economic modes

- shared subscription pool;
- per verified result;
- bounded per-use licence;
- one-time licence;
- premium capacity;
- support/service contract;
- enterprise/private registry;
- atomic task bounty;
- security bounty;
- performance bounty;
- maintenance reward;
- upstream dependency share;
- Arena Recipe share;
- Place design/service revenue;
- creator referral attribution;
- cooperative revenue-sharing;
- voluntary Commons support.

13.4 Licence, provenance, attribution, and settlement are separate layers

Paper IX does not treat an open-source licence, provenance record, attribution receipt, or payment obligation as interchangeable. A reference separation is:

SOURCE LICENCE

permission to use / copy / modify / distribute

|

| does not by itself establish payment

v

PROVENANCE

where the work, capability, data, or method came from

|

v

ATTRIBUTION

which contribution materially participated or survived

|

v

ECONOMIC ENTITLEMENT / SETTLEMENT CONTRACT

which verified events create an enforceable or voluntary payment obligation

A capability may use MIT, Apache, GPL-compatible, proprietary, community-controlled, dual-licensed, or other lawful terms according to its manifest. An MIT licence, for example, permits broad reuse but does not itself require downstream revenue sharing. If a Commons participant is to receive downstream compensation, the entitlement must arise from a separate marketplace agreement, cooperative rule, bounty contract, subscription allocation rule, smart-contract settlement where lawful, or other explicit economic instrument.

The Attestation/Provenance DAG can provide evidence used by such a settlement mechanism. It does not manufacture a legal payment obligation merely by recording a dependency. Likewise, payment does not grant governance, truth, certification, publication, or domain authority.

14. Personal Cognitive Capsule and Portable SLM

14.1 Portable object

The personal SLM is not the whole portable identity.

PERSONAL COGNITIVE CAPSULE

|

+-- canonical semantic primitives

+-- concept relationships

+-- episodic provenance

+-- accepted / rejected interpretations

+-- superseded beliefs

+-- optional Personal SLM

+-- disclosure policy

+-- attestation lineage

+-- model translators

+-- integrity / version roots

+-- revocation / recovery

14.2 Semantic codec

person's language

|

v

Personal Cognitive Capsule

meaning / domain / exclusions /

confidence / provenance

|

v

canonical Aura semantic primitive

```

      |
      v
Arena / collaborator / model
and in reverse:
canonical project state
      |
      v
Personal Cognitive Capsule
      |
      v
person-appropriate explanation

```

14.3 Model portability

A staged reference design:

1. canonical state -> minimal text;
2. canonical state -> typed structured context/tool protocol;
3. compatible open model -> learned prefix/soft prompt;
4. model-specific neural adapter;
5. optional learned cross-model latent translator.

Raw hidden states are not assumed portable.

14.4 Candidate memory

```

external observation
      |
source/provenance binding
      |
trust / relevance assessment
      |
contradiction / supersession check
      |
sandbox learning proposal
      |
regression / consistency / policy
      /      \
reject  promote

```

The capsule can therefore answer not only “what do I remember?” but “why did this become durable?”

15. Governed Recursive Harness Evolution

A reference continual loop is:

```

objective
      |
exact-source orientation
      |
typed continuity
      |
verified dependency graph
      |
bounded recursive workers
      |
machine receipts + verifier evidence

```

```

|
experience / failure attribution
|
candidate learned procedure
|
minimal harness-change proposal
|
isolated evaluation + regression + adversarial tests
|
independent promotion gate
+--> reject/archive
+--> canary
+--> promote
+--> rollback
|
Attestation DAG + intellectual provenance

```

Mutable operational surfaces may evolve. Constitutional truth/authority/verifier definitions cannot be silently rewritten by the system they govern.

16. Intent-Native Manifestation Layer

16.1 Canonical truth versus manifestation

```

CANONICAL MODEL
(source / AST / domain state / contracts / evidence)
|
v
Manifestation Adapter
|
v
task-conditioned spatial / visual / textual / voice presentation
|
user interaction
|
v
Interaction Delta
|
v
typed proposal
|
v
verification / canonical-owner disposition
|
v
materialization, if authorized

```

Generative presentation is allowed to improve comprehension but cannot invent authoritative relationships.

16.2 External research convergence

Recent 2026 work independently illustrates two useful patterns:

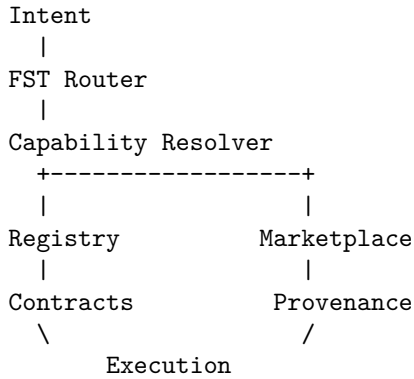
- UniVVT uses semantic conditioning to replace brittle multi-stage inference preprocessing in video virtual try-on.
- MVFGA retains explicit parametric upper-body structure while applying 3D Gaussian appearance for high-fidelity rendering.

Aura's analogous rule is:

structural/canonical truth remains authoritative; rich semantic or Gaussian presentation is an interface.

17. Spatial Code Breadboard

A repository region can manifest as:



Functions become components; interfaces become ports; calls and data flows become wires; tests attach to targets; contracts and security boundaries constrain possible manipulations.

Dragging a new component produces:

```
ghost_component:
  name: NEW_VALIDATOR
  status: NOT_MATERIALIZED
  inputs: [...]
  outputs: [...]
  affected_functions: [...]
  affected_contracts: [...]
  likely_tests: [...]
  provenance: proposal
```

Then:

```
spatial manipulation
|
semantic change request
|
Relational Synthesis
|
minimum relevant source hydration
|
candidate AST/code transformation
|
Surgeon / controlled execution
|
tests + contracts + Verified DAG
|
human/canonical-owner disposition
|
write-back
```

18. Aura Places: Social Media as Governed Spaces

18.1 From profile to Place

The conceptual progression is:

early personal web / MySpace: personalized page
uniform feed platforms: standardized profile
Aura Places: governed portable Place

A Place can include:

- media gallery;
- live studio;
- public room;
- subscriber/backstage room;
- invite-only collaboration room;
- digital closet or product showroom;
- direct commerce;
- creator tools;
- Arena Recipes;
- digital twins;
- customer support;
- repair/returns;
- community events;
- provenance and attribution.

18.2 Place persists, Visit compiles

```
PlaceManifest(version)
|
visitor + objective + relationship
permissions + entitlements
device + network + accessibility
|
v
Ephemeral Visit
|
only missing assets/deltas
|
interaction + receipts
|
v
dissolve temporary state
```

18.3 Relationship-first social architecture

The user can intentionally visit a creator, person, business, or community instead of relying on a compulsory engagement feed.

Supported patterns can include:

- follow/subscribe;
- invite;
- shared Visit;
- public/private rooms;
- chronological discovery;
- user-selected curation;
- direct creator relationships;

- portable relationship edges;
- clear blocking/reporting;
- transparent sponsorship and commerce;
- no conversion of temporary presence into durable personal state without consent.

19. Convention Arena

19.1 Macro-Arena

```

Event objective + governance
    |
    v
admission / exhibitor registry
    |
    v
temporary convention map
+-- central stage
+-- category districts
+-- live rooms
+-- exhibitor Place portals
    |
    v
visitor-specific neighborhood hydration
    |
    v
event interaction / commerce / discovery
    |
    v
deterministic event dissolution

```

The hall dissolves; authorized relationships, receipts, entitlements, subscriptions, purchases, saved artifacts, and digital twins can persist.

19.2 Trend creation

Objective matching is reactive:

“Find a verified red upcycled jacket under \$200.”

Convention discovery can be proactive:

“Walk through the sustainable-fashion innovation convention.”

The second creates a human-governed surface for serendipity, creativity, live drops, and trend formation.

19.3 Example: authenticated cultural convention

A Nation or community may host an art convention whose exhibitor Places are admitted through local identity/provenance rules.

A visitor can inspect a digital representation and the authorized provenance evidence available for that object without receiving restricted cultural knowledge.

20. Open Discovery Foundry and R&D Arenas

20.1 Need-to-deployment lifecycle

```

community / organization need
    |

```

```

jurisdiction + need manifest
|
research objective + prohibited methods
|
local resource characterization
|
literature / data reconstruction
|
capability + simulator composition
|
hypothesis generation
|
multi-model simulation
|
uncertainty / sensitivity / falsification
|
independent review
|
human-approved physical test
|
milestone funding
|
empirical measurement
|
simulation-to-reality comparison
|
repeatability / process capability
|
Commons / cooperative / licensed release
|
fabrication / deployment / repair
|
field learning

```

Maximum creativity is allowed in candidate space. Real-world authority remains narrow.

20.2 R&D assets

The Foundry can treat the following as attributable capability objects:

- datasets;
- literature reconstructions;
- simulation models;
- calibration packs;
- experiment protocols;
- hypothesis generators;
- falsification suites;
- measurement procedures;
- fabrication recipes;
- test fixtures;
- local process adaptations;
- field maintenance improvements.

20.3 Discovery lineage

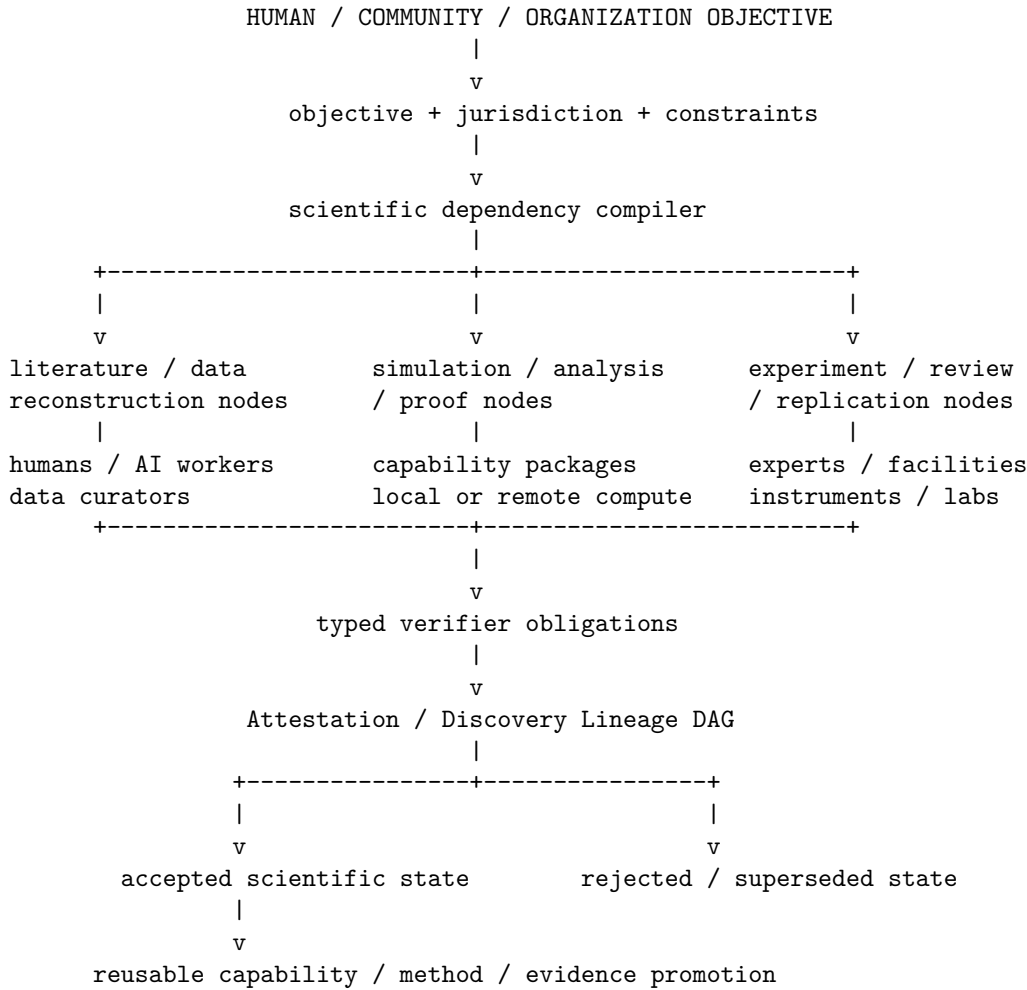
A Discovery Lineage DAG can distinguish:

need originator
 community / organization
 domain expert
 dataset
 paper
 model
 simulation capability
 candidate proposer
 experiment designer
 lab/workshop/operator
 verifier
 fabrication author
 manufacturer
 field maintainer

Not every touched artifact receives equal economic weight.

20.4 Participatory proof-carrying Scientific Arena

The Developer Arena topology generalizes to scientific and engineering work when the work unit changes from a code patch to a bounded scientific obligation. A research objective can be compiled into a dependency graph whose nodes are independently satisfiable by heterogeneous participants:



Illustrative node classes include literature reconstruction, data curation, analytical derivation, parameter sweep, simulator execution, hypothesis generation, adversarial falsification, experiment design, measurement, replication, uncertainty analysis, manufacturing adaptation, and field validation.

A scientific work unit can declare:

```
scientific_work_unit:
  objective_id: ...
  node_id: ...
  truth_class: HYPOTHESIS | SIMULATION_PROJECTION | ANALYTICAL_RESULT | EMPIRICAL_RESULT
  exact_inputs: [...]
  capability_or_method_version: ...
  environment_digest: ...
  resource_budget: ...
  random_seed_or_determinism_contract: ...
  privacy_and_data_rights: ...
  allowed_effects: [...]
  prohibited_effects: [...]
  required_outputs: [...]
  verifier_contract: ...
  upstream_dependencies: [...]
  downstream_consumers: [...]
  attribution_policy_ref: ...
  completion_and_dissolution: ...
```

The node is defined by its scientific contract, not by whether its executor is a person, LLM, SLM, deterministic program, simulator, solver, volunteered computer, paid cloud runtime, laboratory, university, company, community workshop, or other admitted facility. Executor identity and evidence mode remain part of the receipt.

The architecture therefore permits an individual or community with a well-defined problem to originate a research Arena without requiring that one institution own every worker, tool, dataset, simulator, or compute resource. This does not eliminate domain expertise, laboratory cost, certification, physical testing, or regulation. It changes how computational, intellectual, and experimental contributions can be coordinated before consequential action.

20.5 Contributed compute, human participation, and scientific authority

A Scientific Arena may admit voluntarily contributed, purchased, community-owned, enterprise, university, or local edge compute under explicit resource and data contracts. A compute contribution receipt may record:

```
compute_contribution:
  provider_principal: ...
  work_unit: ...
  hardware_class: ...
  runtime_digest: ...
  input_digest: ...
  output_digest: ...
  wall_time: ...
  resource_measurement: ...
  retry_or_replication_state: ...
  verifier_result: ...
  economic_policy_ref: ...
```

Compute expenditure is not scientific truth. A large GPU contribution, high token count, long runtime, financial donation, or large number of agent executions cannot make a hypothesis correct. Scientific acceptance follows the node's verifier contract, evidence class, replication requirements, and where necessary empirical testing.

Likewise, participation may include distinct contribution roles:

```
problem / need originator
scientific objective framer
dataset or evidence curator
method / solver developer
simulation capability maintainer
compute provider
```

hypothesis proposer
 falsification author
 domain reviewer
 experiment designer
 facility / instrument operator
 replication team
 manufacturing or field adapter
 maintenance contributor

The Attestation DAG preserves these roles without assuming equal causal weight. Economic allocation, if any, follows an explicit policy or contract and can distinguish mere execution from a contribution that materially changes the accepted result.

20.6 Compounding Scientific Capability Commons

A reusable scientific improvement can become durable Commons infrastructure only after its own admission and verification. The reference feedback loop is:

```

research objective / unmet need
  |
  v
capability bottleneck or quality gap
  |
  v
open improvement opportunity / bounty / contributor initiative
  |
  v
candidate solver, verifier, dataset, method, adapter, or protocol
  |
  v
benchmark + compatibility + security + scientific validation
  |
  v
Commons promotion as a versioned capability or recipe
  |
  v
rebind into future compatible Scientific Arenas
  |
  v
new execution and outcome evidence
  |
  v
meaningful-use attribution + optional contractual settlement
  |
  v
maintenance / further refinement
  |
  +-----+
  
```

An improved solver does not silently replace every predecessor. Aura may retain multiple implementations for different fidelity, hardware, licence, privacy, jurisdiction, or validation requirements. Routing preference can change only according to current compatibility and evidence. A faster algorithm that fails scientific or numerical validation does not earn authority merely because it reduces cost.

This creates a potential upward development curve in which verified improvements to commonly reused scientific capabilities can reduce the cost or increase the fidelity of later research across many independent objectives while preserving the lineage of the contributors who created, verified, maintained, or materially improved those capabilities.

21. Physics Simulation and Digital Twins

Simulation is explicitly not physical proof.

A high-value candidate can be required to traverse:

```
Simulator A
|
Simulator B / independent approximation
|
analytical checks
|
known benchmark
|
parameter sensitivity
|
uncertainty analysis
|
adversarial falsification
|
qualified empirical test
```

A Simulation Campaign Receipt can include:

```
simulator:
  name: ...
  version: ...
model:
  equation_family: ...
  parameter_digest: ...
boundary_conditions: ...
calibration_dataset: ...
random_seed: ...
uncertainty_method: ...
sensitivity_report: ...
result_digest: ...
known_limitations: ...
empirical_comparison: pending
truth_class: SIMULATION_PROJECTION
```

Applications may include materials, energy, water systems, buildings, construction schedules, logistics, robotics, manufacturing, climate/thermal models, mechanical systems, and other digital-twin domains.

22. People-Behavior and Social Simulation

22.1 Purpose

The architecture can compile exploratory social simulations for:

- emergency planning;
- platform-policy evaluation;
- market/product scenarios;
- public-service design;
- transportation;
- communications;
- community planning;
- business incubation;
- adoption and diffusion hypotheses.

22.2 Safety/validity contract

```
social_simulation:
  population_source: synthetic_or_aggregated
  calibration: declared
  persona_generation: schema_bounded
  models: versioned
  random_seeds: retained
  uncertainty: required
  subgroup_outputs: aggregate_only
  individual_prediction_authority: false
  policy_authority: false
  targeting_authority: false
```

Recent research on LLM-powered social digital twins and interactive policy simulation supports the usefulness of such environments for exploratory counterfactual reasoning. Other work warns that simulated participants require validation before they can support claims about real humans. Aura therefore treats the simulation as a hypothesis surface, not a replacement for empirical evidence or legitimate human governance.

23. Business Incubator Arena

23.1 Demand-to-venture compilation

```
unmet objective / demand signal / Foundry discovery
|
opportunity thesis
|
capability + recipe graph
|
prototype Arena
|
technical / market / behavior / cost simulation
|
risk + regulatory requirements
|
bounty / preorder / grant / milestone proposal
|
human financial/legal disposition
|
pilot
|
measured outcome
|
Place + capability + service + organization
|
continuous improvement / attribution
```

23.2 Demand-backed funding without automatic finance

A community may aggregate privacy-preserving statements that many users want a capability.

For example:

```
1,000 users each indicate willingness to pay $10
|
v
$10,000 conditional development pool
|
```

```

      v
milestone-bound bounty
      |
      v
verified prototype
      |
      v
contributors and initial users receive agreed rights/benefits

```

Aura may compile the opportunity and evidence; it does not autonomously lend, invest, incorporate, sign contracts, issue securities, or spend funds.

24. Sovereign and Cross-Domain Arena Federation

A federation can temporarily compose domains:

```

Health Arena -----+
Housing Arena -----+
Budget Arena -----+--> Temporary Objective Arena
Infrastructure Arena ----+
Environment Arena -----+
Treaty/Governance -----+

```

Each source Arena retains:

- data custody;
- legal/professional/community authority;
- privacy/consent;
- leases;
- verifiers;
- revocation;
- right to disconnect/offline operation.

The higher-order Arena receives minimum authorized projections.

25. Minimized Independent Financial/Compliance Review

For high-accountability workflows, one reference embodiment decomposes a transaction or record into role-specific review units.

```

canonical financial record
      |
purpose-minimizing projector
      |
+-----+-----+-----+
|               |       |
math check    policy  provenance
|             |       |
independent bounded reviewers
+-----+-----+-----+
      |
multi-party disposition
      |
append-only correction / receipt

```

Reviewers do not need unnecessary personal/community identity when it is irrelevant to the check.

This design may reduce opportunities for favoritism, localized pressure, or single-admin abuse, but it does **not** guarantee unbiased reviewers, prevent all collusion, prove source data honest, or automatically satisfy legal/accounting

rules.

26. Convention, Places, and Commons as a New Creative Economy

The economic graph can include multiple contributor roles:

- capability developer;
- optimization developer;
- maintainer;
- recipe creator;
- Place designer;
- Convention organizer;
- verifier;
- security reviewer;
- provenance auditor;
- curator;
- creator/artist;
- domain-rule author;
- R&D model curator;
- experiment designer;
- fabricator;
- repair provider;
- translator/semantic-schema contributor;
- personal/corporate adapter developer.

The architecture can reward specialization without requiring each contributor to own the entire application stack.

27. Continuous Optimization Example

Suppose a Convention Arena consumes too much mobile bandwidth.

demand evidence:

"Convention Visit median unique transfer = 30 MB"

|

v

optimization bounty

|

v

Developer publishes new spatial compressor

|

v

controlled benchmark:

same accepted visual/proof quality,

lower transfer and decode cost

|

v

new package becomes eligible for preferred routing

|

v

meaningful-use receipts record downstream use

|

v

optimizer + legitimate upstream dependencies receive attribution

The older implementation remains available for rollback and compatibility.

28. Reference End-to-End Embodiments

28.1 Proprietary coding capability

A company publishes a private security analyzer.

1. The Commons exposes only its typed contract, rights, evidence mode, version, cost, and output schema.
2. Developer Arena leases a minimal source slice or hash-bound analysis package according to policy.
3. The analyzer executes in an approved private/attested mode.
4. Aura receives findings plus a receipt bound to package/runtime identity.
5. Waboose/canonical source tools independently corroborate high-impact findings.
6. If accepted, the analyzer receives meaningful-use attribution.
7. The company's source stays private.
8. User private source is disclosed only according to the lease.
9. The analyzer gains no patch or merge authority.

28.2 Creator Place and virtual try-on

1. A clothing creator publishes a signed Place and garment assets.
2. A visitor enters from a phone.
3. The Visit loads Commons geometry and only unique creator deltas.
4. With explicit user permission, a virtual-try-on capability receives minimum necessary body/garment representation.
5. The generated view is presentation, not measurement truth.
6. Purchase creates an entitlement and attribution receipt.
7. The creator Place relationship persists.
8. Temporary try-on state dissolves.

28.3 Convention Arena

1. A community publishes event governance and exhibitor-admission rules.
2. Approved exhibitor Place versions enter the event manifest.
3. Each visitor receives a procedural hall neighborhood.
4. Booths portal into exhibitor Places.
5. Talks and launches expose new capabilities, recipes, art, products, or research.
6. Friends can share a Visit/session subject to rights.
7. At event end the hall dissolves.
8. Relationships, purchases, saved artifacts, and permitted receipts remain.

28.4 R&D and business incubation

1. A community posts a need for a lower-cost local building panel.
2. Foundry reconstructs known material data.
3. Physics simulators and analytical checks reproduce known results.
4. Bounded search proposes candidate formulations.
5. Independent falsification rejects weak candidates.
6. Human-approved coupon tests create empirical receipts.
7. A promising design enters the Business Incubator Arena.
8. Demand signals and cost/market simulations support a milestone proposal.
9. Humans decide whether to fund a pilot/cooperative.
10. The resulting fabrication package retains research and contributor lineage.

28.5 Population behavior simulation

1. An emergency-planning objective asks how evacuation communication strategies might affect route demand.
2. The Arena constructs a synthetic calibrated population without representing named residents as digital clones.
3. Several communication scenarios run with reproducible seeds.

4. Experts inspect aggregate outcomes and atypical agent cases.
5. The system highlights uncertainty and assumptions.
6. Results support tabletop planning only.
7. Real authorities make any operational decision.

28.6 Participatory Scientific Arena and compounding capability reuse

1. An individual, community, company, or research group publishes a bounded objective such as improving a low-cost water-treatment component or locally manufacturable building material.
2. The Scientific Arena compiler decomposes the objective into literature, data, analytical, simulation, falsification, experiment-design, and replication nodes.
3. Admitted human experts and AI workers satisfy reasoning and review nodes; local, purchased, or volunteered compute satisfies eligible simulation nodes; specialized facilities remain separately admitted for physical measurements.
4. Every node is bound to exact inputs, method/capability versions, truth class, resource limits, required outputs, and verifier obligations.
5. Candidate results that fail analytical, simulation, replication, or empirical gates remain rejected or advisory and cannot become canonical scientific truth.
6. A contributor supplies a numerically validated solver improvement that reduces runtime while preserving required accuracy.
7. The improved solver passes compatibility, security, benchmark, and scientific validation and is promoted as a new Commons capability version.
8. Future compatible Scientific Arenas may rebind to that version according to current evidence, cost, rights, and fidelity needs.
9. The Attestation/Discovery Lineage DAG preserves the objective originator, method authors, data contributors, compute providers, reviewers, experiment operators, and downstream dependencies.
10. If an explicit Commons settlement agreement applies, meaningful-use evidence can inform compensation; the source licence and provenance record do not themselves create the payment obligation.
11. Physical deployment, certification, funding, procurement, publication, patents, and other consequential dispositions remain separately governed.

29. Threat Model and Failure Modes

29.1 Capability and supply-chain attacks

- manifest lies about effects;
- version substitution;
- provenance laundering;
- malicious dependency;
- dependency confusion;
- source-escrow mismatch;
- TEE measurement mismatch;
- signed vendor runtime behaves differently from test fixture;
- remote service silently upgrades.

Mitigations include digest binding, mode-specific evidence, sandboxing, network/secret restrictions, revocation, independent behavior tests, reproducible builds where possible, transparency/provenance logs, and fail-closed risk classes.

29.2 Attribution gaming

- artificial dependency insertion;
- circular royalty loops;
- invocation spam;
- sybil contributors;
- benchmark overfit;
- sabotaging competitors;

- hidden self-preferencing;
- duplicate wrapper capabilities.

Economic admission should use meaningful-use evidence and maintain dispute/replay paths.

29.3 Personal-memory attacks

- prompt injection becomes durable memory;
- poisoned content is summarized into trusted state;
- unrelated project memory becomes global;
- personal SLM leaks private context;
- adapter rewrites meaning;
- stale semantic mapping is treated as current.

Candidate-memory admission, provenance closure, deterministic disclosure, scope binding, supersession, and roll-back are required.

29.4 Social/Places attacks

- malicious assets;
- covert tracking;
- dark-pattern engagement ranking;
- relationship-based authority confusion;
- abusive shared Visits;
- entitlement theft;
- impersonated creator Place;
- hidden advertising;
- cultural appropriation or unauthorized disclosure.

Places must retain identity, moderation, rights, provenance, disclosure, and community-governance controls.

29.5 Simulation attacks and errors

- simulator exploitation;
- invalid boundary conditions;
- biased calibration;
- overfit behavioral personas;
- unmodeled physical process;
- LLM agents exhibiting unrealistic homogeneity;
- false precision;
- policy maker mistakes projection for forecast.

Simulation truth classes and empirical validation are mandatory.

29.6 Recursive-harness attacks

- child authority escalation;
- recursion/cost explosion;
- self-removal of evaluator;
- benchmark rewriting;
- failure-history deletion;
- memory poisoning;
- compromised worker falsely counted independent.

Authority monotonicity, immutable evaluator ownership, budgets, termination, independent gates, and receipt lineage mitigate these classes.

29.7 Participatory-science and contributed-compute attacks

- volunteer or remote node returns fabricated work;
- heterogeneous hardware changes numerical behavior;
- falsified compute-resource receipts;
- poisoned scientific capability enters a high-reuse path;
- dataset rights are violated during distributed execution;
- duplicate or sybil contributions inflate attribution;
- contributors collude to approve an invalid result;
- a fast solver is preferred despite unacceptable error;
- a large number of agents creates false confidence;
- an economic incentive rewards quantity of computation rather than validated contribution;
- an external facility claims a measurement that cannot be reproduced or traced.

Mitigations include replicated execution where useful, exact environment and input digests, numerical tolerances, independent verifier lineage, hardware/fidelity declarations, data-rights enforcement, benchmark suites, adversarial falsification, empirical rechecking, sybil/dispute controls, and separation between compute credit, economic entitlement, and scientific truth.

30. Defensive Prior-Art Declarations N51-N87

The following declarations are **combination-scoped enabling disclosures**. They do not claim the underlying standard primitives in isolation. Where a claim describes a downstream program not yet deployed at the repository baseline, it is disclosed as a reference embodiment sufficient to establish the system architecture and its negative authority boundaries.

Claim N51: Objective-Native Minimum-Sufficient Arena Compiler

We disclose an objective-native computing substrate in which a human, organization, or community expresses a desired outcome together with constraints, prohibitions, privacy rules, budget, authority, evidence requirements, device constraints, completion criteria, and dissolution criteria; the system then compiles the smallest truthful and sufficient working environment needed for that objective. The compiled environment may be spatial, two-dimensional, voice-first, headless, collaborative, or multimodal and may bind software capabilities, humans, AI workers, data, rule packs, simulations, and services. The combination is distinguished from ordinary application launch, workflow automation, or generative UI by preserving canonical owners outside the temporary environment, applying hard admission before advisory ranking, carrying explicit proof obligations, and requiring deterministic dissolution after completion, denial, expiry, or failure. The Arena cannot create authority absent from its source contracts.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N52: Hierarchical Minimum-Sufficient Evidence Hydration and Semantic Zoom

We disclose a retrieval and manifestation architecture that begins with a compact orientation layer and progressively hydrates exact evidence only at decision points. A request may route from objective to semantic/topological cluster, capability or domain object, symbol or policy region, and finally exact source lines, frames, records, parameters, or authoritative text. The same hierarchy may drive human-facing semantic zoom: high-level architectural objects are presented first; functions, contracts, tests, parameters, or source are revealed only when required. Compact vectors, summaries, topology, VSA/HDC pointers, caches, and projections remain navigation evidence and cannot become patch, domain, or publication authority. The disclosed combination couples computational context reduction and interactive manifestation to the same source-first evidence hierarchy.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N53: Persistent Capability Package With Rights, Provenance, Verification, and Dissolution Contract

We disclose a persistent, versioned Capability Package as the reusable economic and technical unit beneath temporary Arenas. A package declares typed functions and interfaces, inputs, outputs, effects, required permissions, privacy class, resource/cost profile, deterministic or stochastic behavior, dependencies, source/build provenance, licence, verification evidence, compatibility range, limitations, revocation state, cleanup behavior, publisher, maintainers, and execution modes. Selection, purchase, reputation, or popularity does not grant ambient authority. A package is activated only through an objective-scoped lease and must return bounded outputs and closure evidence. This supports public, cooperative, proprietary, organization-private, and community-governed capabilities under one compositional contract family without requiring every implementation to share the same disclosure model.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N54: Arena Recipe as Rebindable Composition Knowledge

We disclose an Arena Recipe that preserves reusable composition knowledge separately from the capabilities it references. A recipe declares an objective class, required functional roles, dependency graph, proof obligations, fallback logic, interaction or projection pattern, dissolution rules, and optional economic attribution terms. At execution time the recipe is rebound to current capability versions, licences, permissions, evidence, device state, budgets, policies, and canonical owners; it therefore cannot serve as a frozen second planner or authority source. Recipes may be free, paid, private, community-governed, institution-certified, or collaboratively maintained. The same recipe abstraction can compose coding, research, commerce, education, civic, construction, convention, and other Arenas while retaining exact runtime rebinding and verification.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N55: Ephemeral Activation, Explicit Promotion, and Deterministic Dissolution

We disclose a lifecycle in which persistent capabilities, recipes, identities, licences, and approved evidence are activated into short-lived objective-specific state, and temporary outputs become durable only through an explicit promotion gate. Transient context, leases, renderer state, proposal branches, temporary secrets, worker reasoning state, caches, and generated layouts are destroyed or released at closure. A successful adapter, test, recipe, learned primitive, or artifact may be promoted only with source identity, proof, provenance, policy, and canonical-owner disposition. Dissolution emits evidence of what was released, retained, unavailable, or failed to clean up, and emergency termination cannot fabricate cleanup. The combination prevents temporary application state from silently accumulating durable authority.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N56: Federated Aura Commons With Plural Registries and Anti-Capture Portability

We disclose a Commons architecture supporting interoperable global, private, enterprise, professional, cooperative, community, and Nation-governed registries. The Commons coordinates discovery, portable manifests, capability and asset admission, provenance, licences, entitlements, verification, payment references, receipts, accessibility, moderation, appeals, migration, and federation while leaving local governance and canonical domain authority with their owners. Packages, Places, receipts, entitlements, and dependency identities are portable; registries may disconnect, revoke, migrate, or operate offline without granting a global registry ownership of local data or culturally restricted knowledge. Ranking is separated from authority and may be user-configurable. This combination treats federation and anti-monopoly portability as constitutional properties of a capability economy rather than optional export features.

Negative authority boundary: This declaration does not claim the standard primitives named within it in

isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N57: Executable Rights and Entitlements as Effect-Bounding Objects

We disclose an executable-rights representation that binds a principal, resource, permitted effect, purpose, scope, duration, transferability, delegation policy, revocation path, evidence requirement, appeal path, and constitutional ceiling. Rights may cover access, publish, delegate, revoke, modify, distribute, subscribe, purchase, receive attribution or payment, operate a capability for a bounded period, use community-governed knowledge, or fabricate under a specified licence. Relationships, payment, credentials, popularity, or ownership labels do not automatically imply transitive authority. An Arena must resolve the applicable right before opening an effectful action, and the right cannot grant more authority than its issuer possesses. This provides an explicit bridge between capability security, Commons entitlements, and real-world governance.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N58: Proprietary Capability Participation Without Mandatory Source Disclosure

We disclose a mechanism by which proprietary code can participate in Aura’s Ephemeral Arena and Commons model without surrendering source. A minimized public or counterparty manifest exposes contracts, effects, permissions, identity, licence/entitlement, compatibility, evidence class, and verification requirements while implementation bytes may remain within an organization-private runtime, encrypted sandbox, trusted execution environment, source-escrow verifier, signed vendor runtime, or remote service. The system distinguishes evidence modes such as `OPEN_SOURCE_REPRODUCIBLE`, `SOURCE_ESCROW_REPRODUCIBLE`, `TEE_ATTESTED`, `SIGNED_VENDOR_RUNTIME`, and `REMOTE_SERVICE_ONLY` rather than pretending they provide identical assurance. Meaningful-use receipts can support attribution, accounting, revocation, and downstream provenance without exposing proprietary source or private user objectives. High-risk domains may require stronger modes before admission.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N59: Semantic-Gate Execution Receipts and Branching Attestation DAG

We disclose compact execution receipts emitted at semantically meaningful boundaries rather than per token or internal model operation. Illustrative gates include objective admission, intent normalization, constraint admission, capability selection, graph admission, authority grant, action opening, material state change, verification, result disposition, human/community disposition, and persistence/publication/closure. Receipts bind parent receipts, run/task identity, principal, operation, input/output hashes, intent/constraint hashes, source and implementation identity, authority, environment, verification, result, failure class, evidence pointer, and time. Because work branches and rejoins, receipts form a DAG rather than a single log. Full prompts, source trees, reasoning transcripts, and test logs remain in their canonical stores and are referenced rather than duplicated.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N60: Receipt Frontier and Lazy Provenance Hydration

We disclose a provenance architecture in which active computation retains a compact frontier consisting of current state identity, current receipt, causal parent references, and optional checkpoint/Merkle root. Historical receipts, evidence, source, or supporting events are recursively hydrated only when a current task needs an explanation, failure ancestry, licence closure, attribution analysis, rollback, or audit. This preserves reconstructability without recreating long-context bloat. Retrieval follows provenance and authority classes: a summary cannot upgrade an upstream claim; a derived lesson retains origin; and rollback identifies affected descendants. The same sparse execution memory can orient new workers and support failure analysis, learning, and accounting.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N61: Machine Execution Attestation Separated From Human/Institutional Responsibility Attestation

We disclose an accountability stack that treats machine execution evidence and human or institutional declarations as different epistemic objects. A machine receipt may establish that a specified agent or tool produced output from identified inputs under a named policy and that a verifier reported a defined result. A human or institution may separately attest that it reviewed a particular artifact or claim and stands behind a bounded declaration at a moment in time. Hashes prove byte relationships; signatures prove key action; tests prove only their covered predicates; neither machine receipts nor human declarations alone prove universal truth. The architecture allows an external human-accountability mechanism, including an Opus-Facio-like attestation service, to bind responsibility above Aura’s machine provenance without becoming Aura’s verifier, truth owner, or authorization plane. Specific external attestation algorithms are not claimed here.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N62: Durable External Agent Identity Bound to Internal Authority and Historical Action Lineage

We disclose an identity composition in which a durable external accountable-agent identifier, including a DNSid-style DNS-anchored FQDN identity, is treated as an edge identity anchor while Aura separately binds its internal principal, objective, delegated authority lease, runtime instance, model/provider identity, execution receipt, and artifact lineage. Identity continuity can therefore survive operational-key rotation, worker termination, model replacement, Arena dissolution, or provider migration without treating a new runtime as a new source of authority. The external identity anchor answers which accountable entity stands behind an agent; Aura answers what that principal was authorized to do and what occurred. The external identity system is not used as an automatic trust, reputation, policy, or execution grant.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N63: Meaningful-Use Attribution From Verified Contribution Graphs

We disclose contribution attribution computed from proof-carrying execution and dependency evidence rather than raw invocation, token, message, branch, or commit count. A contribution record may incorporate whether a capability or work unit executed, whether its output survived integration, whether downstream nodes consumed it, verifier disposition, failure prevention, supersession, user/canonical-owner acceptance, upstream dependencies, maintenance/security role, cost/latency, provenance quality, and counterfactual or hindsight estimates of marginal contribution. Attribution remains contestable, auditable, and repairable when later evidence changes the causal picture. This creates an economic substrate in which code, verification, data, recipes, translations, design, maintenance, scientific models, fabrication processes, and human review can receive distinct credit.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N64: Evidence-Driven Continuous Optimization Economy

We disclose a feedback economy in which objective demand and measured bottlenecks create bounties or opportunities; developers or specialists publish candidate optimizations; the harness benchmarks them under comparable proof obligations; routing or composition preferences may shift only after verified improvement; and meaningful-use attribution can direct future economic share toward the improved capability and its legitimate upstream dependencies. Improvements may target latency, compute, bandwidth, memory, energy, accessibility, security,

compatibility, quality, or maintenance. The system archives prior champions and rejected candidates rather than overwriting lineage. Economic reward is therefore coupled to evidence-backed contribution to successful objectives, creating an incentive for continuous refinement of shared primitives without granting the winning package permanent or exclusive authority.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N65: Developer Arena as Proof-Carrying Heterogeneous Work-Unit Fabric

We disclose a Developer Arena in which the unit of collaboration is an Atomic Work Unit, not a chat message or unrestricted shared branch. Each unit binds objective, canonical semantics, exact source/state identity, allowed mutation scope, dependencies, interface contract, positive requirements, explicit prohibitions, authority, budget, proof obligations, expected outputs, failure classes, retry/reuse rules, parent hashes, attribution terms, and completion receipt requirements. Human developers, LLMs, personal SLMs, static analyzers, fuzzers, test generators, profilers, formal solvers, security scanners, documentation agents, and maintainers may satisfy different nodes. Workers receive only minimum-sufficient context and return proposal/evidence state; canonical owners admit verified results.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N66: Critical-Path Compiler and Contract-First Parallelization

We disclose a planning/execution compiler whose objective is to minimize verified wall-clock completion latency rather than maximize agent count. It identifies genuinely independent work, hidden shared state, interfaces that must freeze first, tasks executable against contracts or mocks, verification that can proceed concurrently, invalidation rules, and the current critical path. It recursively converts avoidably sequential work into independently verifiable concurrent work by stabilizing interfaces before implementations complete. The compiler respects Amdahl-type limits and does not claim linear speedup from worker multiplication. Parallel work is admitted only when dependency and authority evidence supports independence.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N67: Bounded Speculative Branches With Deterministic Reintegration and Partial Re-Execution

We disclose controlled speculative execution in which multiple candidate implementations, plans, simulations, or hypotheses may run concurrently when uncertainty would otherwise block progress and when cost, safety, and objective value justify the fan-out. Each branch has an exact base, authority ceiling, budget, provenance, verifier, and dissolution path. A judge or canonical owner may admit one, combine compatible evidence, reject all, or retain a branch as an archive. Upstream changes produce typed invalidation so unaffected verified work can be reused while dependent nodes are replayed. No speculative branch gains production authority merely by finishing first or scoring well on its own evaluator.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N68: Reviewer Independence and Heterogeneous Verification Lineage

We disclose verification evidence that records not only reviewer conclusions but also independence-relevant lineage: model family/provider, prompt/transcript exposure, source acquisition route, retrieval independence, inherited findings, shared mutable state, correlated tools, and human/organizational relationship where relevant. Multiple agreeing reviewers are not counted as independent proof merely because they are separate processes. A verification

swarm can deliberately combine different models, deterministic tools, humans, formal methods, and source-retrieval paths. This counters correlated multi-agent conformity and supports stronger proof classes for high-impact work.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N69: Execution Traceability Separated From Causal Credit Assignment

We disclose a two-layer treatment of provenance: the Attestation DAG establishes what events and dependencies occurred, while a separate causal analysis may estimate which upstream transitions materially caused a success or failure. Causal analysis can use dependency structure, failure localization, controlled replay, ablation, counterfactual substitution, and hindsight credit methods. Its outputs remain labeled estimates unless directly proven by a bounded counterfactual test. This distinction prevents deterministic routing or chronological ancestry from being misrepresented as causal contribution, while still allowing economic attribution and debugging to become more precise over time.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N70: Personal Cognitive Capsule With Canonical Semantics, Provenance, Personal SLM, Policy, and Adapters

We disclose a user-owned Personal Cognitive Capsule that persists independently of any one foundation-model provider. The capsule includes canonical semantic primitives, concept relationships, episodic provenance, accepted and rejected interpretations, superseded beliefs, optional personal SLM or learned neural memory, disclosure policy, attestation lineage explaining why durable primitives exist, model-specific translators, integrity/version roots, and revocation/recovery state. The personal SLM is a component, not the portable source of truth. Durable personal learning remains versioned, provenance-linked, scope-bounded, and subject to promotion rules. The capsule provides continuity without turning a model vendor’s hidden state into the user’s identity.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N71: Personal-to-Canonical Semantic Codec and Collaboration Interlingua

We disclose a bidirectional semantic translation layer that maps a person’s idiolect, definitions, cultural/domain references, abstraction preferences, positive objectives, negative objectives, and known exclusions into provider-neutral canonical Aura primitives, and maps canonical project state back into person-appropriate explanations. Canonical representation may use typed concept IDs, relations, scopes, VSA/HDC bindings, confidence, provenance, supersession, and authority state. Different contributor phrases that resolve to the same canonical primitive can be deduplicated while preserving original wording and contribution identity. Translation is explicitly an interpretation, not truth or authority, and material uncertainty can trigger clarification or constrained alternatives.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N72: Personal Cognition as Minimum-Disclosure Privacy and Memory-Security Membrane

We disclose a privacy architecture in which the full private cognitive history remains behind the Personal Cognitive Capsule and external Arenas, models, Places, or contributors receive only the minimum sufficient semantic projection authorized for requester, purpose, domain, duration, and sensitivity class. A learned personal model may rank relevance or propose disclosure, but deterministic policy and cryptographic controls decide. External information enters as candidate memory: it is source-bound, trust/relevance assessed, checked for contradiction and supersession, evaluated in a sandbox, and promoted only after appropriate consistency and governance gates. Economic attribution references contribution receipts rather than exposing the underlying private cognitive record.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N73: Provider-Neutral Portable Personal AI With Progressive Model Adapters

We disclose model-independent personal continuity in which portable canonical semantics and provenance can be projected into multiple host models through progressively richer adapters: minimal text projection, typed structured context/tool protocol, soft-prompt or prefix adapters for compatible open models, model-specific neural adapters, and optional learned cross-model latent translators where architectures permit. Raw hidden states or weights are not assumed universally portable. Adapters are versioned, tested, revocable behavioral artifacts and cannot become authority owners. A user may change foundation models while retaining the canonical personal cognitive record, its provenance, policy, and learned semantic history.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N74: Recursive Programmability and Continual Harness Learning Without Recursive Authority

We disclose a recursively programmable worker substrate in which persistent sessions and child workers may refine methods, supplemental prompts, skills, decomposition strategies, routing hints, or other explicitly mutable operational surfaces while immutable or separately governed constitutional surfaces retain control over truth classes, authority ceilings, verification definitions, retention, publication, promotion, and merge. Child authority is monotone non-increasing relative to its parent. Candidate harness changes predict their intended effects, execute in isolation, undergo regression/adversarial evaluation, and move through reject, archive, canary, promote, or roll-back states under an independent gate. Execution/evolution receipts preserve causal and intellectual provenance. The program, not the LLM, owns admission and stopping.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N75: Intent-Native Manifestation Adapter Over Canonical Truth

We disclose a domain-generic manifestation architecture that projects canonical state into a task-conditioned semantic, spatial, visual, textual, voice, or generative representation without transferring canonical ownership to the representation. A Manifestation Adapter declares the canonical schema it reads, projection schema, allowed affordances, interaction-delta compiler, verifier, provenance/uncertainty display, lazy-hydration hooks, and materialization protocol. User interactions produce proposal deltas that are translated back into typed change requests and independently verified before any canonical state change. Rich generative appearance may be used for comprehension, while exact source, contracts, domain records, or structural models remain truth.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N76: Spatial Code Breadboard as a Proposal-Only Programming Front End

We disclose a coding manifestation in which a verified repository region is rendered as an interactive breadboard: functions or services appear as components, interfaces as ports, calls/data flows as wires, contracts and security boundaries as constrained regions, tests as attached proof targets, and provenance/ownership as metadata. Semantic zoom hydrates exact source only when needed. Dragging, wiring, grouping, or requesting a new component creates a ghost/proposed structure with inferred interfaces, affected symbols, contracts, tests, and candidate regressions; it does not mutate source. The proposal is translated through relational synthesis to exact source hydration, candidate AST/code transformation, controlled Surgeon execution, tests/contracts/Verified DAG, and authorized write-back.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N77: Aura Places: Persistent Governed Place, Ephemeral Visitor-Specific Visit

We disclose Aura Places as persistent signed, versioned, portable definitions owned or governed by a person, creator, business, organization, community, or Nation, while each Visit is an ephemeral compilation for one visitor, relationship, objective, permission set, entitlement set, device, network condition, accessibility need, and context. A Visit resolves the signed Place version, checks rights, compares required assets to local availability, selects the smallest adequate representation, transfers only missing deltas, records minimized proof/attribution, and dissolves temporary runtime state while applying user retention choices. The Place manifest and rights are canonical; rendered scenes, caches, presence, and generated layout are not.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N78: Relationship-First Social, Creator, and Commerce Network of Custom Governed Places

We disclose a social architecture that replaces the uniform platform profile/feed as the primary human surface with user- or organization-authored governed Places. A Place may combine identity/story, media, live rooms, public/private/invited/subscriber zones, creator workflows, virtual try-on or digital twins, direct commerce, subscriptions, commissions, customer support, repair/returns, loyalty, co-design, referrals, provenance, and Arena Recipe distribution. Social discovery emphasizes intentional relationships, explicit invitations, user-controlled/chronological discovery options, blocking/reporting, portable relationship records, and bounded shared Visits rather than compulsory engagement-ranking. This preserves the customization spirit of early personal web pages while combining it with signed manifests, objective-native compilation, rights, provenance, and ephemeral delivery.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N79: Convention Arena as an Objective-Conditioned Macro-Arena of Places

We disclose a Convention Arena that compiles an event from an event objective, governance/admission rules, authenticated or vetted exhibitor registry, schedules, stages, category districts, Commons primitives, and participant Places. The hall is not a permanently running monolithic world: only the visitor's relevant spatial neighborhood and unique assets are hydrated, and exhibitor booths act as portals into signed Place versions. Conventions can support live drops of capabilities, Arena Recipes, research discoveries, creative works, products, talks, and collaborative rooms. When the event closes, the temporary hall and presence state dissolve while authorized relationships, purchases, entitlements, attribution receipts, digital twins, subscriptions, and event artifacts may persist. Community or Nation hosts retain their own admission and cultural rules.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N80: Dual Discovery: Reactive Objective Matching Plus Proactive Human-Governed Trend Discovery

We disclose a discovery system that deliberately preserves two modes. Reactive intent matching resolves an already expressed objective against compatible capabilities, products, services, Places, or evidence. Proactive discovery occurs through user-chosen Conventions, curators, communities, subscriptions, exploratory Places, or thematic showcases that expose people to ideas they did not know to request. The proactive mode is not permitted to silently convert behavioral surveillance into authority or covert manipulation; ranking provenance, sponsorship/commerce relationships, and user controls remain visible. This combination allows an objective-native system to respond

efficiently to existing demand without suppressing novelty, creative trend formation, or serendipitous human discovery.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N81: Open Discovery Foundry: Governed Need-to-R&D-to-Deployment Lifecycle

We disclose an Open Discovery Foundry that compiles a community, organization, or individual need into a bounded R&D lifecycle: need/jurisdiction manifest; research objective and prohibited-method contract; local-resource characterization; literature/data reconstruction; capability and simulation-plan composition; hypothesis generation; multi-model simulation and analytical checks; uncertainty/sensitivity analysis; adversarial falsification; independent review; human-approved physical experiment; staged milestone funding; empirical measurement; simulation-to-reality comparison; repeatability/process-capability testing; Commons/cooperative/licensed release; local fabrication/deployment; repair and field learning. Open-ended creativity is allowed in candidate space, while funding, procurement, publication, equipment control, certification, and external actuation remain separately authorized.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N82: Multi-Simulator Physics and Digital-Twin Evidence Stack With Simulation/Reality Separation

We disclose a simulation evidence architecture for physics, engineering, construction, energy, materials, infrastructure, and other physical domains in which a candidate is evaluated across identified simulator versions, parameter sets, boundary conditions, analytical sanity checks, benchmark cases, sensitivity analysis, uncertainty propagation, adversarial falsification, and, when necessary, independent simulator or approximation families before physical validation. A Simulation Campaign Receipt binds model identity, assumptions, parameters, calibration data, outputs, failure modes, and comparison evidence. Simulation remains a screening/projection truth class and cannot become sensor truth, certification, survey truth, engineering approval, or physical proof. Empirical measurements are compared back to simulations to update model confidence rather than retrospectively hiding disagreement.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N83: Bounded Social and Population Behavior Simulation as Advisory Counterfactual Arena

We disclose a people-behavior simulation Arena that creates privacy-minimized synthetic or statistically representative populations and schema-constrained agent models for exploratory counterfactual analysis of policy, product, market, social, emergency, or community scenarios. Each campaign binds population construction method, calibration data, uncertainty, sampling assumptions, model/provider identities, scenario inputs, outcome distributions, subgroup aggregation rules, and known validity limits. Individual simulated agents are not treated as predictions about identified persons; synthetic responses cannot authorize resource allocation, policing, healthcare, benefits, governance, or individualized targeting. Results are advisory hypotheses to be compared against empirical data, expert judgment, and where appropriate real human research. Branching scenarios preserve assumptions and reproducible seeds/configuration.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N84: Business Incubator and Demand-Backed Venture Formation Arena

We disclose a Business Incubator Arena that converts unmet objectives, privacy-preserving demand signals, or validated research outputs into bounded venture-development workflows. The system can compile opportunity theses,

technical capability graphs, Arena Recipes, prototype Arenas, competitor/substitute analysis, cost/resource models, market and behavior simulations, regulatory/risk requirements, milestone bounties, group preorders or demand commitments, pilot criteria, verification, contributor rights, and candidate organizational forms such as business, cooperative, community enterprise, or service network. Human and institutional principals retain investment, contract, incorporation, procurement, hiring, lending, and equity authority. Successful ventures may publish a Place, capabilities, recipes, services, or fabrication packages whose downstream usage remains attributable to contributing people and dependencies.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N85: Sovereign Federated Cross-Domain Arenas With Minimized Independent Review and Multi-Party Disposition

We disclose temporary federation of independently governed Arenas in which each domain, organization, community, or Nation retains its own data custody, legal/professional/cultural authority, identity/privacy/consent rules, capability leases, verifier gates, and right to revoke or operate offline. A higher-order objective Arena receives minimum authorized projections and exact references rather than unrestricted underlying records. For accountability-sensitive work such as financial review, procurement, grants, or compliance, tasks may be decomposed into minimized or pseudonymized review units and routed to independent reviewers under deterministic checks, while multi-party/threshold disposition and append-only correction reduce single-point abuse, favoritism, or collusion risk. The architecture does not claim that anonymity eliminates bias, that cryptography guarantees honest source data, or that the mechanism automatically satisfies any jurisdiction’s legal requirements.

Negative authority boundary: This declaration does not claim the standard primitives named within it in isolation, and the disclosed component cannot silently become a canonical truth, authority, verification, identity, payment, or publication owner beyond its explicit contract.

Claim N86: Participatory Proof-Carrying Scientific Arena With Heterogeneous Contributors

We disclose a Scientific Arena in which a human, community, organization, or other authorized originator expresses a bounded research or engineering objective and the system compiles that objective into a typed scientific dependency graph. Graph nodes may represent literature reconstruction, data curation, analytical derivation, simulation, parameter search, hypothesis generation, falsification, experiment design, measurement, replication, uncertainty analysis, fabrication adaptation, or field validation and may be satisfied by admitted humans, AI workers, deterministic programs, open or proprietary capability packages, local/remote/volunteered compute, laboratories, instruments, workshops, or other authorized facilities. Each node binds exact inputs, method/capability version, environment identity, resource limits, privacy/data rights, truth class, required outputs, verifier contract, upstream dependencies, downstream consumers, and completion/dissolution state. Execution, resource expenditure, agent count, consensus, or economic contribution cannot by themselves promote a hypothesis or simulation into scientific truth; promotion follows the declared evidence and verification path, including empirical validation where required. The Attestation/Discovery Lineage DAG preserves heterogeneous contribution roles and enables partial invalidation, replay, replication, and current reproof without requiring one institution to own every contributor or resource.

Negative authority boundary: This declaration does not claim volunteer computing, multi-agent science, distributed workflows, scientific simulation, citizen science, laboratories, or provenance systems in isolation. A participant, compute provider, model, simulator, or facility cannot gain scientific, funding, publication, certification, procurement, patent, governance, or physical-actuation authority merely by executing a node or contributing resources.

Claim N87: Compounding Scientific Capability Commons With Verified Reuse and Explicit Settlement

We disclose a feedback architecture in which bottlenecks, quality gaps, cost constraints, or new research objectives generate improvement opportunities for scientific capability packages, datasets, verifiers, solvers, simulators, methods, experiment protocols, adapters, or Arena Recipes; candidate improvements are independently benchmarked for compatibility, security, numerical/scientific validity, evidence quality, and declared domain constraints before

durable promotion; promoted versions may then be rebound into later compatible Scientific Arenas according to current objective, rights, fidelity, jurisdiction, cost, privacy, and evidence requirements; subsequent execution produces new outcome and provenance evidence; and meaningful-use/counterfactual contribution analysis can update reputation, routing preference, maintenance demand, bounties, or contractual economic settlement. Multiple versions may coexist, and a faster or more popular capability cannot supersede a predecessor solely on cost, invocation count, compute expenditure, or model preference. Source licence, provenance, attribution, and payment are separate layers: a licence grants stated use rights, the Attestation DAG records lineage, attribution estimates contribution, and any downstream compensation requires an explicit economic policy or agreement.

Negative authority boundary: This declaration does not claim open-source licensing, scientific marketplaces, volunteer-compute credit, citation, bounties, or software royalties in isolation. Reuse evidence cannot manufacture scientific truth, legal entitlement, intellectual-property ownership, or payment authority beyond the applicable contract, and economic incentives cannot override scientific verification or community/domain governance.

31. Claim-to-Embodiment Crosswalk

- **N51-N55 - Objective-native compilation and lifecycle:** hierarchical hydration, Capability Packages, Arena Recipes, explicit promotion, and deterministic dissolution. Representative surfaces include the Spatial/Ephemeral refactor contracts and reference Arena compiler.
- **N56-N58 - Commons, rights, and proprietary participation:** plural public/private/Nation registries, executable entitlements, evidence-classed proprietary runtimes, and source-confidential contribution receipts.
- **N59-N64 - Identity, attestation, provenance, and economics:** semantic-gate receipts, receipt-frontier hydration, DNSid-bound external accountability, machine/human responsibility separation, meaningful-use attribution, and optimization economics.
- **N65-N69 - Developer Arena:** Atomic Work Units, critical-path compilation, bounded speculation, reviewer-independence lineage, and counterfactual causal analysis.
- **N70-N74 - Personal cognition and continual harness:** canonical personal semantics, candidate memory, Personal SLM, model adapters, privacy policy, and recursively programmable learning without recursive authority.
- **N75-N76 - Manifestation and code breadboard:** source-grounded projections, ghost components, typed interaction deltas, selective source hydration, and verified materialization.
- **N77-N80 - Places, social/commerce, Convention, and discovery:** persistent Place plus ephemeral Visit, creator/community rooms, Convention macro-Arenas, and user-governed reactive/proactive discovery.
- **N81-N84 - Foundry, simulation, and incubation:** multi-simulator R&D campaigns, human-approved empirical tests, bounded synthetic-population counterfactuals, and demand-backed venture pilots.
- **N85 - Sovereign federation and minimized independent review:** temporary cross-domain objective Arenas, minimum authorized projections, decomposed financial/compliance work, and multi-party disposition.
- **N86-N87 - Participatory science and compounding scientific capabilities:** heterogeneous proof-carrying Scientific Work Units, contributed compute/facilities, scientific truth-class separation, verified reusable capability promotion, and explicit separation of licence, provenance, attribution, and settlement.

32. Related Work and Explicitly Non-Claimed Primitives

The following adjacent systems and standards demonstrate relevant primitives or independent convergence. Paper IX does not claim them in isolation.

32.1 Identity, credentials, signatures, and provenance

- **DNS-Anchored Durable Identity for AI Agents (DNSid), draft-ihsanullah-dnsid-01, IETF Internet-Draft, 29 June 2026.** Durable accountable-agent identity anchored to DNS/FQDN, lifecycle history, and keys. Aura's disclosure begins where that external identity is bound to objective-specific authority, execution lineage, and contribution.
- **W3C Verifiable Credentials Data Model 2.0, Recommendation, 15 May 2025.** Machine-verifiable credential representation. Aura may consume credentials as evidence but does not treat possession as automatic authority.

- **Sigstore / Fulcio / Rekor.** Identity-associated software signing and transparency logging. Aura’s Attestation DAG and execution semantics extend beyond artifact signing into objective, authority, state transitions, verification, attribution, and dissolution.
- **in-toto, SLSA, TUF, SPDX, W3C PROV.** Important supply-chain/provenance precedents. Aura combines provenance with ephemeral objective execution, canonical ownership, rights, and economic attribution.

32.2 Federated social systems

- **ActivityPub (W3C Recommendation, 2018).** Decentralized social networking protocol.
- **AT Protocol.** Decentralized identity, signed personal data repositories, schema-based Lexicons, federation, and account migration.
- Aura Places does not claim decentralized social identity or federation in isolation. Its combination is persistent governed Place manifests plus visitor-specific ephemeral compilation, objective/rights conditioning, lazy asset deltas, capability/recipe composition, deterministic dissolution, and proof/attribution economics.

32.3 Agent harnesses and verification

- **Prime Intellect verifiers v1 (2026).** Separates tasksets, harnesses, and runtimes for modern agentic evaluation.
- **Prime Intellect’s prime infrastructure.** Provides sandboxes, environments, evaluation, compute, and training infrastructure.
- Aura’s distinct combination couples replaceable/recursive workers to exact source, authority monotonicity, execution receipts, canonical owners, persistent provenance, governed learning, and human disposition.

32.4 Spatial/semantic manifestation

- **UniVVT (Cao et al., arXiv:2608.05745, 2026).** Semantically conditioned video try-on reducing reliance on explicit intermediate preprocessing at inference.
- **MVFGA (Javanmardi et al., arXiv:2608.04722, 2026).** Parametric structural mesh combined with dynamic 3D Gaussian rendering.
- Paper IX uses these as independent support for semantic conditioning and structure/presentation separation, not as claimed Aura primitives.

32.5 Social and policy simulation

- **LLM-Powered Social Digital Twins (Gupta & Sheikh, arXiv:2601.06111, 2026).**
- **LLM-Augmented Digital Twin for Policy Evaluation in Short-Video Platforms (Zhang et al., arXiv:2603.11333, 2026).**
- **WhatIf: Interactive Exploration of LLM-Powered Social Simulations for Policy Reasoning (Li et al., arXiv:2604.17615, 2026).**
- **Hullman et al., “This human study did not involve human subjects” (arXiv:2602.15785, 2026).** Relevant warning about the conditions under which LLM simulations can support behavioral evidence.
- Paper IX therefore claims neither LLM agents nor population simulations in isolation; it discloses their bounded integration into Aura’s objective, evidence, authority, provenance, privacy, and empirical-validation architecture.

32.6 Multi-agent science and volunteer computing

- **Zhang et al., The Virtual Biotech: A Multi-Agent AI Framework for Therapeutic Discovery and Development (bioRxiv, 2026).** The Stanford-led system mirrors a therapeutic R&D organization with a Chief Scientific Officer agent and specialized scientist agents. In one demonstration it dispatched 37,075 clinical-trialist agents in parallel and analyzed outcomes from 55,984 clinical trials. This is strong prior art for large-scale domain-specific agent decomposition and tool-using scientific analysis. Paper IX does not claim that pattern in isolation.
- **Ghareeb et al., A multi-agent system for automating scientific discovery (Nature, 2026).** Robin integrates literature search, hypothesis generation, experiment proposal, experimental-data analysis, inter-

pretation, and updated hypothesis formation in a structured multi-agent biology workflow. This is prior art for multi-stage agentic scientific discovery.

- **BOINC / BOINC Central.** Volunteer computing predates Aura and enables scientific jobs to run on contributed consumer CPU/GPU resources; BOINC Central is explicitly intended to broaden high-throughput volunteer computing access, including to independent researchers. This is prior art for public contributed compute and distributed scientific execution.
- **Aura’s disclosed combination begins at a different integration boundary:** an authorized objective is compiled into heterogeneous proof-carrying scientific work nodes that may be satisfied by humans, agents, reusable capabilities, contributed compute, and facilities; scientific truth classes and independent verification remain explicit; accepted reusable improvements can be promoted into a federated capability Commons; lineage survives downstream reuse; and any economic allocation is separately governed by meaningful-use evidence plus explicit settlement terms rather than raw compute credit, agent count, citation, or source licence alone.

33. Why the Combined Architecture Differs From a Conventional App Store

A conventional app store generally resolves:

user chooses app -> app runs

The disclosed Aura model resolves:

objective
-> canonical semantics
-> minimum evidence
-> required functional roles
-> current rights and policy
-> compatible capability candidates
-> verified composition
-> ephemeral Arena
-> bounded human/machine work
-> proof
-> canonical-owner disposition
-> attribution/economic settlement
-> dissolution

The economic network can therefore reward reusable specialist components and composition knowledge rather than requiring every provider to own a complete monolithic application.

34. Why the Combined Architecture Differs From Conventional Social Media

Conventional social systems generally persist content/account state and render it through standardized apps.

Aura Places adds:

portable governed Place definition
+
visitor relationship / objective / rights
+
device / accessibility / bandwidth state
+
Commons capabilities / assets / recipes
=
ephemeral Visit

The Place can function as the creator’s or community’s durable digital home, while each visit remains a bounded compilation rather than one global persistent virtual world.

35. Why the Combined Architecture Differs From Conventional Multi-Agent Systems

A conventional multi-agent orchestration system may route tasks among agents.

Paper IX adds combination constraints:

- exact source/state identity;
- bilateral intent;
- authority leases;
- canonical ownership;
- proof-carrying Atomic Work Units;
- critical-path decomposition;
- heterogeneous reviewer independence evidence;
- semantic-gate receipts;
- lazy provenance hydration;
- causal attribution separate from traceability;
- meaningful-use economics;
- personal semantic translation/privacy;
- deterministic dissolution;
- recursive learning without recursive authority.

36. Why the Combined Architecture Differs From a Blockchain-Only Attribution System

A ledger can establish order, signatures, and tamper evidence. It does not by itself know whether:

- a capability was actually used;
- its result survived integration;
- verification passed;
- it was causally important;
- the input was truthful;
- a human reviewed the result;
- a payment is legally owed;
- a community permitted the use.

Paper IX therefore treats blockchains, Merkle structures, transparency logs, or append-only databases as possible persistence mechanisms beneath a richer typed Attestation/Attribution DAG, not as the source of semantic truth.

37. Governance and Cultural Authority

The Commons can support Nation/community registries and collective rights.

The architecture must be capable of representing:

- individual authority;
- family/collective authority;
- community/Nation authority;
- professional authority;
- organizational authority;
- territorial/jurisdictional rules;
- cultural restrictions;
- benefit-sharing;

- purpose limitation;
- revocation;
- return of findings.

Personal consent does not automatically override collective governance where the relevant governing system requires collective authority.

Not all knowledge must be globally discoverable.

38. Safety and Negative Authority Invariants

The following are normative design constraints of the disclosed reference architecture:

1. No second truth plane.
2. No second memory plane.
3. No second planner/router.
4. No second verifier.
5. No second identity or payment truth plane.
6. Presentation is not authority.
7. A model answer is not evidence by itself.
8. Consensus is not independent verification.
9. Simulation is not physical proof.
10. Synthetic people are not real-person predictions.
11. A relationship is not transitive authority.
12. A purchased capability gets no ambient authority.
13. A signed artifact may still be unsafe or false.
14. A Place manifest can be canonical while its rendering is not.
15. A Place persists; a Visit dissolves.
16. Capabilities persist; activations dissolve.
17. Every effectful operation retains a canonical owner.
18. Every proprietary execution mode has an explicit evidence class.
19. Every durable learned primitive has provenance and supersession semantics.
20. Personal SLM output cannot self-authorize disclosure.
21. Receipts should minimize disclosure.
22. Observability must not become uncontrolled surveillance.
23. Payment follows meaningful verified contribution, not traffic volume alone.
24. FST/WFST traceability does not by itself establish causal credit.
25. Massive worker count does not imply linear speedup.
26. Work not proven independent is not executed as if independent.
27. Recursive workers cannot acquire authority their parents lack.
28. Harness learning cannot rewrite its own constitutional evaluator without independent governance.
29. Autonomous R&D cannot autonomously control funding, procurement, certification, publication, or equipment.
30. Social/behavior simulation cannot become individual targeting or allocation authority.
31. Minimized/blind review reduces some bias/collusion surfaces but cannot guarantee neutrality.
32. Cryptographic provenance does not prove the original factual assertion was true.
33. External identity proves an accountable binding, not trustworthiness or authorization.
34. Human responsibility attestation proves a declaration, not universal truth.
35. Deterministic dissolution is tested and receipted.

39. Implementation and Demonstration Sequence

This defensive publication does not require the entire system to be deployed simultaneously.

A disciplined implementation sequence is:

Current Spatial / Ephemeral foundation (PR2-PR8)

|

```

      v
Capability Package contracts (PR9)
|
supply-chain / sandbox / revocation (PR10)
|
semantic compatibility + composition planner (PR11)
|
first-party registry / Developer Arena pilot (PR12)
|
meaningful-use attribution simulation (PR13)
|
invited third-party capabilities (PR14)
|
Commons governance + private/community registries (PR15)
|
limited premium marketplace (PR16)
|
federated registries + portable provenance (PR17)
|
cross-domain stabilization / external review (PR18)
|
+---> Places program
+---> Convention program
+---> Personal Capsule maturity
+---> Discovery Foundry / simulation
+---> Participatory Scientific Arena / contributed-compute pilot
+---> Business Incubator

```

Every future phase should rerun Aura’s current exact-head Harness, Connectome, Atlas, Compass, Relational Synthesis, and canonical-owner analysis rather than blindly implementing old file paths.

40. Evaluation Metrics

40.1 Objective compilation

- success under explicit objective/constraints;
- minimum-sufficient evidence volume;
- stale/unknown rejection;
- composition digest stability;
- partial re-execution correctness;
- deterministic dissolution.

40.2 Developer Arena

- wall-clock completion;
- critical-path length;
- useful parallel fraction;
- coordination overhead;
- invalidation rate;
- integration conflict;
- regression rate;
- independent reproduction;
- context hydration volume.

40.3 Provenance

- receipt bytes/task;

- lineage reconstruction;
- omission detection;
- lazy-hydration rate;
- rollback/causal-closure accuracy.

40.4 Economics

- payout explainability;
- dependency gaming;
- sybil/fraud rate;
- upstream recognition;
- maintenance incentive effectiveness;
- disputes and corrections;
- user cost transparency.

40.5 Personal Capsule

- semantic translation accuracy;
- duplicate-concept detection;
- disclosure minimization;
- stale-belief handling;
- candidate-memory poisoning resistance;
- cross-model portability;
- provenance retrieval.

40.6 Places/Convention

- Place manifest size;
- cold/warm/delta bytes;
- time to meaningful Visit;
- local cache reuse/reclamation;
- entitlement recovery;
- provenance retention;
- accessibility parity;
- event dissolution correctness;
- user-controlled discovery.

40.7 Foundry/simulation/incubation

- known-result reproduction;
- simulator calibration;
- model disagreement detection;
- uncertainty quality;
- falsification success;
- simulation-to-reality error;
- repeatability/process capability;
- scientific dependency-graph completion;
- heterogeneous executor reproduction rate;
- contributed-compute receipt integrity;
- verifier rejection and false-acceptance rate;
- promoted capability reuse across independent objectives;
- measured cost/fidelity improvement after capability upgrades;
- stale or incompatible capability rejection;
- attribution and settlement explainability by contributor role;
- demand-signal validity;
- pilot conversion;
- contributor-lineage accuracy.

41. Limitations and Open Research

The architecture does not establish:

- that all capability types are safely composable;
- that causal contribution can always be computed fairly;
- that economic allocation will be sybil-proof;
- that proprietary execution can achieve open-source-level inspectability;
- that trusted execution environments are free of side channels;
- that a Personal SLM can faithfully model a person;
- that latent model adapters will be portable;
- that Places will outperform existing social networks;
- that Convention Arenas will create healthy trends;
- that simulation predicts real physical or human systems;
- that R&D outputs will pass certification;
- that every research objective can be safely or efficiently decomposed into independent work units;
- that volunteered or remote compute is trustworthy, numerically identical, continuously available, or appropriate for sensitive data;
- that causal scientific contribution can always be converted into a fair economic share;
- that broad participation removes the need for qualified experts, laboratories, replication, peer review, or regulatory oversight;
- that demand-backed funding avoids securities or consumer-protection law;
- that minimized auditing eliminates bias or collusion;
- that any proposed financial/governance workflow satisfies Canadian or other law;
- that DNSid becomes an IETF standard;
- that an external attestation service provides legal liability in every jurisdiction.

Those are future engineering, scientific, legal, economic, security, and governance questions.

42. Publication and Reproducibility Record

For defensive-publication use, a publication package should preserve:

1. this paper in PDF and source form;
2. the exact AuraOS repository baseline SHA;
3. Paper VIII reference;
4. the current master refactor plan;
5. the Future Updates architecture addendum;
6. the Objective-Native / Commons / Places / Foundry synthesis;
7. ARCH v2.2;
8. a dated source bundle or repository tag where practical;
9. checksums of the publication files;
10. a public timestamped repository/Zenodo record.

A patent professional can advise on jurisdiction-specific defensive-publication and patent strategy. This technical record is designed to make the combinations and embodiments explicit, not to substitute for legal counsel.

43. Suggested Metadata

Title: AuraOS Paper IX: Objective-Native Capability Commons and Proof-Carrying Contribution Economies

Subtitle: Portable Cognitive Capsules, Proprietary Capability Execution, Developer Arenas, Attestation Lineage, Aura Places, Convention Arenas, Participatory R&D, Simulation, and Business Incubation

Claim range: N51-N87

Suggested categories: cs.AI; cs.SE; cs.DC; cs.HC; cs.CY; cs.MA

Keywords: objective-native computing; ephemeral applications; capability marketplace; capability security; provenance; attestation; agent identity; Personal Cognitive Capsule; portable SLM; developer arena; participatory scientific arena; volunteer computing; scientific workflow; capability commons; social computing; spatial computing; digital twins; simulation; research automation; business incubation; Indigenous data governance; federated systems; defensive publication.

44. Conclusion

Paper VIII established an evidence-ordered cognitive operating substrate that could discover broadly while acting narrowly.

Paper IX discloses what follows when that substrate is treated as the foundation of a larger computing economy.

The permanent unit is no longer necessarily the application.

It can be:

a verified, attributable, rights-bearing capability that can be composed into temporary objective-specific Arenas.

The personal unit is not a vendor-owned chat history.

It can be:

a portable, provenance-bearing cognitive capsule that translates a person's meaning into shared canonical semantics while retaining privacy and model independence.

The collaboration unit is not a chat thread.

It can be:

an independently verifiable work node in a proof-carrying dependency graph.

The social unit is not merely a standardized profile in an engagement feed.

It can be:

a persistent governed Place whose Visits compile ephemerally around each relationship, objective, device, and permission state.

The research unit is not an unconstrained autonomous scientist.

It can be:

a governed proof-carrying Foundry in which a research objective can be decomposed among humans, AI workers, reusable scientific capabilities, contributed compute, and authorized facilities while keeping scientific truth, funding, publication, physical action, certification, and community authority outside raw participation or autonomous model control.

The economic unit is not raw attention or invocation count.

It can be:

meaningful verified contribution to a successful objective, with provenance preserved through reuse, adaptation, maintenance, and downstream composition.

The resulting architecture is a possible transition from app-centric computing toward a **governed Commons of persistent capabilities, portable personal cognition, temporary environments, verifiable contributions, customizable Places, participatory Scientific Arenas, bounded simulations, compounding reusable research infrastructure, and locally actionable knowledge.**

Its strongest design rule remains the same as Aura's earlier architecture:

Intelligence may propose, compose, simulate, visualize, and learn broadly. Authority, truth, verification, and consequential action remain explicit, bounded, attributable, and independently governable.

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